

BUS 123 · MATH-M04-L01 · Markup, Markdown & Break-Even

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M04 · **Lesson:** L01 **Case Study Company:** Anchor & Oak Events

1 · CONNECT TO PRIOR KNOWLEDGE

In the discounts lesson, you started with a list price and reduced it. In this lesson, you often **start with cost and build up to a selling price**. The percentage mechanics are familiar, but the business question changes: will this price cover the cost, reflect the risk, and still make sense to the customer?

2 · CORE CONCEPTS

Part A — Markup

Markup is the amount added to cost to create a selling price. Anchor & Oak Events may know the ingredient and packaging cost for a tasting box, but that cost is not the price. The price must also cover labor, event preparation, waste, and profit.

A 45% markup on cost means the selling price equals cost plus 45% of cost. In a spreadsheet, this is modeled by placing cost in one cell, markup rate in another, and writing a formula that multiplies cost by one plus the markup rate.

Selling Price = Cost × (1 + Markup Rate)

Markup on Cost vs. Markup on Selling Price

A common source of confusion is **the base**. The same dollar amount of markup produces a different percentage depending on what you divide by.

Measure	Formula	Example (\$18 cost, \$26.10 price)
Markup Amount	Selling Price – Cost	\$26.10 – \$18.00 = \$8.10
Markup on Cost	Markup Amount / Cost	\$8.10 / \$18.00 = 45%
Markup on Selling Price	Markup Amount / Selling Price	\$8.10 / \$26.10 = 31%

Same dollars — different percentage. Whenever you read a business problem, **pause and identify the base** before choosing a formula.

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PRE-READING

Part B — Markdowns and Perishables

Markdowns reduce price after the original price has been set. For an event business, markdowns can help fill late seats or sell perishable items before they lose value. **A markdown is not automatically a failure** — it is a tradeoff: lower margin per sale in exchange for more cash recovery, fuller attendance, or less waste.

Markdown Price = Original Price × (1 – Markdown Rate)

Waste Changes the Pricing Problem

Perishable products require extra care because not every produced unit may become a sellable unit. If Anchor & Oak produces 120 tasting boxes and expects 15% waste, only 102 boxes are expected to be sold.

Key Insight

The cost of all 120 boxes still needs to be recovered from the 102 sellable boxes. This is why a price that looks profitable before waste can become too low after waste is considered.

Sellable Units = Produced Units × (1 – Waste Rate)

Part C — Break-Even

Break-even is the point where **total revenue equals total cost**. Before break-even, the business is losing money on the event. After break-even, each additional unit contributes toward profit.

The key idea is **contribution margin**: selling price minus variable cost per unit. Contribution margin is not the same as profit, because fixed costs still need to be covered first.

Contribution Margin = Selling Price – Variable Cost per Unit

Break-Even Units = Fixed Costs ÷ Contribution Margin per Unit

The Rounding Rule

Break-even units must always be rounded up to the nearest whole unit. Selling to 171 guests when the break-even is 171.43 would still leave a small loss. Use `=ROUNDUP(result, 0)` in Excel — never round down.

3 · WORKED EXAMPLES

Worked Example 1 — Markup With Waste

Anchor & Oak plans to produce 120 tasting boxes. Each box costs \$18 to prepare, and the target markup is 45% on cost. The business expects 15% waste.

Step 1 — Target Revenue (assuming all boxes sell):

- $120 \times \$18.00 \times 1.45 = \$3,132$

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Step 2 — Sellable Units:

- $120 \times (1 - 0.15) = \mathbf{102 \text{ boxes}}$

Step 3 — Required Price per Sellable Box:

- $\$3,132 \div 102 = \mathbf{\$30.71}$

Notice what happened: the price rose because fewer sellable boxes must recover the same production cost.

`` Excel pattern: Target Revenue = =ProducedUnits \* CostPerUnit \* (1 + MarkupRate) Required Price = =TargetRevenue / SellableUnits ``

Worked Example 2 — Break-Even

Anchor & Oak is considering a tasting event with \$3,600 in fixed costs, a \$48 ticket price, and \$27 variable cost per guest.

Step 1 — Contribution Margin:

- $\$48 - \$27 = \mathbf{\$21 \text{ per guest}}$

Step 2 — Break-Even Guests:

- $\$3,600 \div \$21 = 171.43 \rightarrow \text{round up to } \mathbf{172 \text{ guests}}$

`` Excel pattern: =ROUNDUP(FixedCost / (TicketPrice - VariableCost), 0) ``

4 · FORMULA REFERENCE

Concept	Formula
Markup Amount	Selling Price – Cost
Markup on Cost	Markup Amount / Cost
Selling Price (markup)	Cost × (1 + Markup Rate)
Markdown Price	Original Price × (1 – Markdown Rate)
Sellable Units	Produced Units × (1 – Waste Rate)
Contribution Margin	Selling Price – Variable Cost per Unit
Break-Even Units	Fixed Costs / Contribution Margin per Unit

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Concept	Formula
Break-Even Revenue	Break-Even Units × Selling Price

5 · CHECK YOUR UNDERSTANDING

Answer each question before looking at the answer key below. Show your formula setup, not just the final number.

- 1 A dessert box costs \$20 and is marked up 40% on cost. What is the selling price?
- 2 A \$50 ticket is marked down 20%. What is the new price?
- 3 Anchor & Oak produces 160 favors and expects 10% waste. How many are sellable?
- 4 An event has a \$35 ticket price and \$18 variable cost per guest. What is the contribution margin?
- 5 Fixed cost is \$2,400 and contribution margin is \$16. How many units are needed to break even?

Answer Key · Check Your Understanding

#	Answer
1	$\$20 \times (1 + 0.40) = \$20 \times 1.40 = \mathbf{\$28.00}$
2	$\$50 \times (1 - 0.20) = \$50 \times 0.80 = \mathbf{\$40.00}$
3	$160 \times (1 - 0.10) = 160 \times 0.90 = \mathbf{144 \text{ favors}}$
4	$\$35 - \$18 = \mathbf{\$17 \text{ per guest}}$
5	$\$2,400 / \$16 = \mathbf{150 \text{ units}}$ (exact — no rounding needed here)

6 · KEY VOCABULARY

Term	Definition
Markup	The amount added to cost to create a selling price.
Markup on Cost	Markup measured as a percentage of cost. Formula: Markup Amount / Cost.

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PRE-READING

Term	Definition
Markup on Selling Price	Markup measured as a percentage of the selling price. Formula: $\text{Markup Amount} / \text{Selling Price}$.
Markdown	A reduction from an original selling price. Formula: $\text{Original Price} \times (1 - \text{Markdown Rate})$.
Perishable Inventory	Goods or capacity that lose value over time and cannot be held indefinitely for sale.
Waste Rate	The percentage of produced units expected to be unsellable. Reduces the denominator when calculating required price.
Variable Cost	Cost that changes with each additional unit produced or guest served.
Fixed Cost	Cost that does not change with production volume in the short run (e.g., venue rental, permits).
Contribution Margin	Selling price minus variable cost per unit. The amount each unit contributes toward covering fixed costs and profit.
Break-Even Point	The sales volume at which total revenue exactly equals total cost; zero profit, zero loss.

Bring to Class

You will use the starter workbook to test the same pricing logic from this reading. Expect short Live You Try It pauses for markup, markdown, and break-even formulas before moving into the Class Challenge. The challenge is where you combine the formulas and explain whether Anchor & Oak's pricing decision is financially reasonable.